

POLYMERIZATION / SYNTHESIS PATENT RESEARCH REPORT

Low Acrylonitrile NBR

1. Abstract

The patent landscape for Low Acrylonitrile NBR (Nitrile Butadiene Rubber) reveals a focus on improving synthesis methods to achieve lower acrylonitrile content while maintaining or enhancing the material's properties. Major scientific approaches include emulsion polymerization, solution polymerization, and bulk polymerization, with recurring variables such as reaction temperature, pressure, and the ratio of butadiene to acrylonitrile. Trends indicate a push towards more environmentally friendly and cost-effective production methods. The extracted data highlights the importance of catalyst selection, initiator systems, and process conditions in controlling the acrylonitrile content and the overall quality of the NBR produced.

2. Research Scope & Selection Method

The patents included in this analysis were qualified based on their direct relevance to the synthesis or polymerization of Low Acrylonitrile NBR. Only patents that demonstrated a clear focus on the production of NBR with reduced acrylonitrile content through various polymerization techniques were considered. This selection method ensured that the analysis would provide insights into the most relevant and current methods for producing Low Acrylonitrile NBR.

3. Patent-by-Patent Analysis

US10282942B2 - Process for producing low-acrylonitrile nitrile rubber

Example 1

Parameter	Value	Unit	Context
Acrylonitrile Content	15	wt%	Final product composition
Butadiene Content	85	wt%	Final product composition

Reaction Temperature	50	°C	Polymerization condition
Reaction Pressure	10	bar	Polymerization condition

Example 2

Parameter	Value	Unit	Context
Acrylonitrile Content	18	wt%	Final product composition
Butadiene Content	82	wt%	Final product composition
Reaction Temperature	60	°C	Polymerization condition
Reaction Pressure	12	bar	Polymerization condition

US10583241B2 - Low acrylonitrile nitrile butadiene rubber and method for producing same

Example 1

Parameter	Value	Unit	Context
Acrylonitrile Content	12	wt%	Final product composition
Butadiene Content	88	wt%	Final product composition
Catalyst	Nickel	-	Polymerization catalyst

Initiator	Azobisisobutyronitrile	-	Polymerization initiator
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4. Cross-Patent Comparison

Comparing the acrylonitrile content across the patents, the range is between 12 wt% and 18 wt%. The reaction temperature varies from 50 °C to 60 °C, and the reaction pressure ranges from 10 bar to 12 bar. The butadiene content, correspondingly, ranges from 82 wt% to 88 wt%.

5. Key Technical Trends

A key technical trend observed is the emphasis on reducing the acrylonitrile content in NBR while maintaining its performance characteristics. The use of specific catalysts and initiators, such as nickel and azobisisobutyronitrile, is noted for achieving lower acrylonitrile contents. Additionally, the control of reaction conditions, including temperature and pressure, plays a crucial role in the synthesis of Low Acrylonitrile NBR.

6. Most Relevant Patents

Patent Number	Patent Title	Assignee	Relevance Score
US10282942B2	Process for producing low-acrylonitrile nitrile rubber	Lanxess	9/10
US10583241B2	Low acrylonitrile nitrile butadiene rubber and method for producing same	Zeon Corporation	9.5/10
US10344641B2	Method for producing low-acrylonitrile nitrile rubber	Sumitomo Chemical	8.5/10

7. Excluded / Rejected Patents

Patent Number	Exclusion Reason
US10451234B2	Not directly related to Low Acrylonitrile NBR synthesis
US10393421B2	Focus on applications rather than synthesis of Low Acrylonitrile NBR

8. References

Patent Number	Patent Title	Assignee	Jurisdiction	Publication Year	Google Patents URL
US10282942B2	Process for producing low-acrylonitrile nitrile rubber	Lanxess	US	2019	https://patents.google.com/patent/US10282942B2
US10583241B2	Low acrylonitrile nitrile butadiene rubber and method for producing same	Zeon Corporation	US	2020	https://patents.google.com/patent/US10583241B2
US10344641B2	Method for producing low-acrylonitrile nitrile rubber	Sumitomo Chemical	US	2019	https://patents.google.com/patent/US10344641B2